SYDE572 – Assignment 2

Linear Regression

In this question, you will implement linear basis function regression with polynomial and Gaussian basis.

The dataset is Housing dataset from the UCI repository. The task is to predict median house value from features describing a town. UCI no longer hosts this dataset has there were concerns of racial bias. However the dataset can still be accessed from the following website
https://www.cs.toronto.edu/~delve/data/boston/bostonDetail.html

Part 1: Polynomial Basis Function
Implement linear basis function regression with polynomial basis functions.
Use only monomials of a single variable $(x_1, x_1^2, x_1^3, \ldots, x_2, x_2^2, \ldots)$ and no cross-terms (x_1, x_2).

a) Create a python function polynomial regression for the following:

Using the first 100 points as training data, and the remainder as testing data, fit a polynomial basis function regression for degree 1 to degree 7 polynomials. Do not use any regularization. Plot training error and test error (in RMS error) versus polynomial degree.

Put this plot, along with a brief comment on what you see, in your report.

b) Run your polynomial regression using a degree 1 polynomial. Examine the learned weights. What value is chosen for w_5, the weight on the 6th feature (average number of rooms per dwelling)? What value is chosen for the weight on the 8th feature (weighted distance to five Boston employment centers)? (Don't forget the bias weight and the normalizations.) Do these 2 weights seem reasonable?
Put the values of all weights and your comments on weights for the 6th and 8th features in your report. You do not need to submit code for this part.

c) Create a python function polynomial-regression-1d-vis for the following:
It is difficult to visualize the results of high-dimensional regression. Instead, only use one of the features (use $X_n(:,2)$) and again perform polynomial regression. Produce plots of the training data points, learned polynomial, and test data points. Do not forget the normalization.

Put 3 of these plots, for interesting (low-order, high-order polynomials) results, in your report. Include brief comments.

d) Create a python polynomial-regression-reg function for the following:

Implement L2-regularized regression using the first 100 points, and only the 2nd feature. Fit a degree 8 polynomial using each value in {0; 0:01; 0:1; 1; 10; 100; 1000} for λ. Use 10-fold cross-validation to decide on the best value of λ. Produce a plot of average validation set error versus the regularizing constant . Use a semilogx plot, putting the regularizing constant λ on a log scale. The unregularized result (λ = 0) will not appear on this scale. You can either add it as a separate horizontal line as a baseline, or report this number separately.

Put this plot in your report, and note which regularizing constant λ you would choose for the cross-validation.

Aside: for reference on cross-validation, use the following tutorial https://quantdev.ssri.psu.edu/tutorials/cross-validation-tutorial

Part 2: Gaussian Basis Function

Implement linear basis function regression with Gaussian basis functions. For the centers j use randomly chosen training data points.

e) Create a python function gaussian-regression for the following:

Use the first 100 points as training data, and the remainder as testing data. Fit a Gaussian basis function regression using 5; 15; 25; ; 95 basis functions (Generate a random permutation of points and pick the first K points as the center of basis functions). Do not use any regularization. Plot the training error and test error (in RMS error) versus number of basis functions.

Put this plot, along with a brief comment on what you see, in your report.

f) Create a python function gaussian-regression-reg for the following:

Implement L2-regularized regression. Again, use the first 100 points (do not only use the second feature, use them all). Fit a regression model with 90 basis functions using each value in {0; 0:01; 0:1; 1; 10; 100; 1000} for λ. Use 10-fold cross-validation to decide on the best value of λ. Produce a plot of average validation set error versus the regularizing constant λ. Use a semilogx plot, putting the regularizing constant λ on the log scale (see note previously about λ= 0).

Put this plot on your report, and note which regularizing constant you would choose from the cross-validation.

Grading Rubric

This assignment will be graded out of **15** points, as follows:

For analytical results: /10. (A+ is 9.5)

- **A. (8-10)**
- **B (7-8)**
- **C (6-7)**
- **D (5)**
- **F (3)**

For presentation of data: \5

- **4.5 excellent**
- **4 good quality**
- **3 average**
- **2 below average**

Use guidelines for assignment 1 for submission